Product Vision & Positioning

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Revision: 2 **Last modified:** 2026-06-26T12:00:00Z

> **Document role.** This is the Volume 1 (Product & Requirements) deep document that fixes *why HelixVPN exists, what category it competes in, who came before, and what makes it different*. It deepens §1/§2/§6/§7 of the volume overview `00-product-scope-and-principles.md` and §1/§2 of the spine `SPECIFICATION.md`. It is the narrative spine the functional requirements (`functional-requirements.md`) and the persona model (`personas-and-roles.md`) trace back to.
>
> **Status:** SPEC-ONLY — this describes the product's *position*, it does not build it. Every market or competitor *claim* that is not an architectural fact of HelixVPN is marked `UNVERIFIED` (§11.4.6): comparative-product facts move over time and are subject to the §11.4.99 latest-source re-verification pass (tracked in `REFINEMENT_NOTES.md` R1). Architectural claims about HelixVPN itself are cited to the overview/spine, not asserted from memory.
>
> **Evidence base.** The synthesised overview and spine (themselves a synthesis of the 16 source research docs), cited inline as `[00 §N]` (the overview) and `[SPEC §N]` (the spine). Where the overview cites a primary source by id (`[04_ARCH §N]`, `[05_YBO]`, `[research-masque]`) that chain is preserved.

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1. Vision statement

HelixVPN is a self-hostable overlay network with a privacy-VPN front end — one codebase that a single person can stand up for a homelab and that the same images scale to an HA multi-region fleet [00 §1], [SPEC §1].

The one-sentence pitch, verbatim from the architecture source [00 §1] (`[04_ARCH §1]`):

> *Cloudflare Tunnel + WARP, rebuilt as Tailscale-style coordination, with Mullvad's obfuscation stack, fully self-hostable, on one shared codebase.*

The vision is deliberately a **union**, not a new point product. Four mature lineages each solved one piece; no shipping product unifies all four *and* is self-hostable [00 §1]:

Lineage	The one thing it proved	HelixVPN inherits
Cloudflare Tunnel + WARP	a connector can **dial outbound** so no inbound port-forward is ever needed; MASQUE/QUIC is a viable censorship-resistant transport	the outbound-only edge model + MASQUE transport `[04_ARCH §0/§3.3]`
Tailscale / Headscale	coordination as a **pushed desired-state network map** (ACL-compiled topology), not a polling mesh	the push-based control plane `[04_ARCH §4.4]`
Mullvad	a privacy bar — full obfuscation stack, anonymous accounts, no-logging — is a *product*, not a research demo	the obfuscation + privacy bar `[04_ARCH §6]`
NetBird	the closest OSS shape (WireGuard + management/ signal server, self-hosted) as a sanity check	architectural validation of the WG-core + control-server split `[04_ARCH §1.3]`

The synthesis target: a product that is **self-hosted AND Mullvad-grade obfuscated AND multi-network-routed AND reachable on 8 platforms from one codebase** — the combination the overview names as the core differentiator [00 §6], [SPEC §2].

2. The problem HelixVPN exists to solve

2.1 The founding constraint

The project was born from one concrete operator need [00 §1] (`[05_YBO]`, `[00]`):

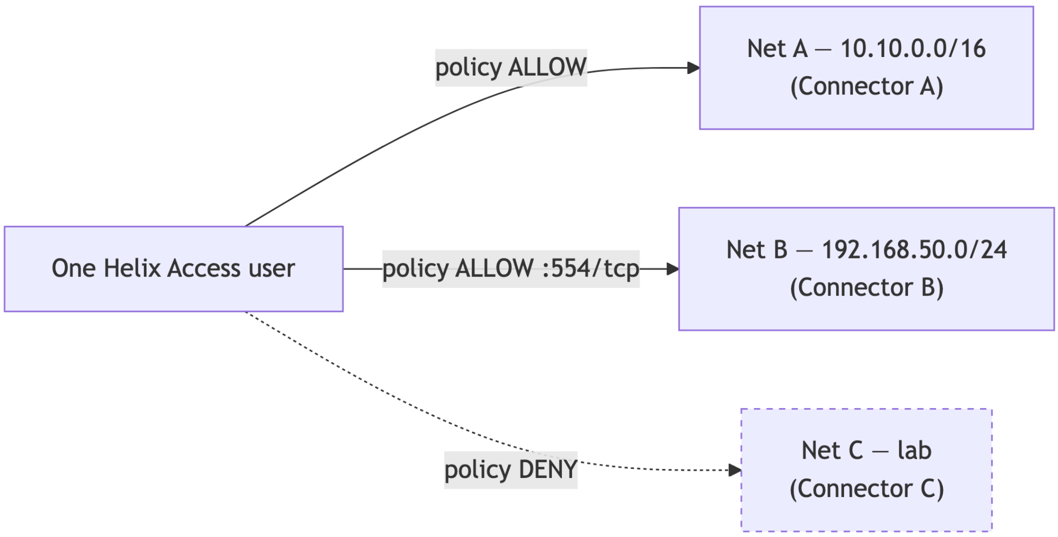
> A remote user must obtain full, **policy-scoped** access to one *or many* internal / home / lab networks **without any inbound port-forward** on those networks.

Every incumbent forces a trade here: you either open a router port (a public inbound attack surface), or you accept a SaaS coordination dependency, or you give up obfuscation, or you give up self-hosting. HelixVPN's thesis is that **none of those trades is necessary** — internal hosts dial *outbound* to a public gateway (a reverse tunnel), the gateway relays and routes, and the whole thing is software you run yourself [00 §2.1], [SPEC §1].

2.2 The headline capability gap in the market

The differentiator the overview leads with is **`1 user → N joined private networks`** — a multi-network, bidirectional, policy-routed gateway, which the overview states "no self-hostable product ships today" [00 §1] (`[04_ARCH §1.3]`).

> **UNVERIFIED** — "no self-hostable product ships today" is a competitive-market claim, not an architectural fact of HelixVPN. It is plausible from the prior-art map (§4) but MUST be re-verified against current product feature sets in the §11.4.99 latest-source pass before it appears in any external-facing material (R1). The *architectural* fact — that HelixVPN's design provides 1→N policy-routed networks — is cited and is not UNVERIFIED.



The user reaches the *subset of joined networks its ACL grants* and is denied the rest — default-deny microsegmentation, not all-or-nothing access [00 §4.2].

2.3 The privacy problem — credibility of "no-logs"

A privacy VPN's no-logs promise is only as strong as your trust in the operator. HelixVPN's structural answer: **no-logging is a build property you can verify, not a policy you must trust** — there is no durable connection/traffic/packet table in the schema, and CI schema-lint fails the build if one appears [00 §8 P7], [SPEC §4.7]. For a self-hoster the claim is *self-evident* (you own the box); for a future managed SKU it is the *contractual* promise backed by the same CI-enforced schema [00 §7.3]. This is detailed in `../v05-security/no-logging-as-code.md` and the security overview `../04-security-privacy-pki.md` (S6/S7).

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3. Market category — the consumer/business convergence

HelixVPN sits at a convergence point of two categories that have historically been separate products [00 §6], [SPEC §2]:

Category	Canonical examples	What HelixVPN takes
Consumer privacy VPN	Mullvad, IVPN, ProtonVPN	obfuscation stack, anonymous accounts, kill-switch, no-logging, one-button connect
Business overlay		

Category	Canonical examples	What HelixVPN takes
network / ZTNA	Tailscale, NetBird, Twingate, Zscaler, Cloudflare Access	coordination/network-map, per-user ACL routing, device posture, multi-tenancy, audit

The same Gateway serves **both** use cases from one deployment [00 §3.3]:

1. **Privacy-exit mode** (the Mullvad use case) — the Client full-tunnels to the internet through the Gateway; the Gateway is a plain privacy exit.
2. **Overlay-access mode** (the ZTNA use case) — the Client reaches the subset of joined private networks its policy allows, via Connectors that dialed out.

The convergence is not a marketing framing; it falls out of the architecture. The *same Rust core* captures the device's traffic in both modes; the *same control plane* compiles policy in both modes; the difference is only which exit/route the network map grants [00 §5], [SPEC §3]. This is why a single codebase can address a consumer persona (privacy-consumer) and three business personas (business-admin, business-end-user, connector-operator) — all enumerated in `personas-and-roles.md`.

> **UNVERIFIED** — the strategic claim that "the consumer privacy-VPN and business ZTNA categories are converging" is an industry-trend assertion, not an architectural fact. It is the *rationale* for the dual model, but its market-truth is out of scope for this spec and is not relied on by any requirement; requirements derive from the architecture, not the trend.

3.1 Hosting/commercial stance (settled direction, open licence)

The hosting model is settled: **self-host / home-lab first; the same code can serve managed later** [00 §7.1], [SPEC §2]. The *licensing* model is an **open decision (D8)** — surfaced with options + recommendation, never silently picked [00 §7.2], [SPEC §9]:

- **Recommendation:** permissive (Apache-2.0/MIT) for the reusable cores (`helix-core`, `helix-ui`, `helix-proto`) so they are maximally reusable by other Helix-ecosystem projects (§11.4.28/.74), plus source-available (BSL-class) for the commercial multi-tenant Console/managed layer [00 §7.2].
- **Status:** OWNER-GATED (§11.4.66) before public release; until resolved every repo carries an explicit `LICENSE.PENDING` marker, never an assumed licence (§11.4.6) [00 §7.2].

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4. Prior-art deep-dive

This is the positioning map from [00 §6], expanded with *what HelixVPN borrows, what it deliberately diverges on, and the honesty caveat per competitor*. The comparison rows are competitive-product facts and carry an `UNVERIFIED` blanket caveat (§4.6).

4.1 Mullvad — the obfuscation & privacy bar

- **What it is:** a consumer privacy VPN with a best-in-class obfuscation stack (QUIC/MASQUE, Shadowsocks, UDP-over-TCP, lightweight obfs, DAITA), multi-hop, kill-switch, anonymous account numbers, and a strong no-logging stance [00 §6].
- **What HelixVPN borrows:** the *entire obfuscation + privacy feature set* — it is the acceptance checklist (the Mullvad-parity matrix, [00 §9]). The single most load-bearing inherited insight: **Mullvad's "QUIC mode" is not a separate protocol — it is WireGuard-over-MASQUE/HTTP-3** [00 §2.2], (`[research-masque]`). This shapes the entire data plane (D1).
- **What HelixVPN diverges on:** Mullvad is a *service* you cannot self-host, and it is a *privacy exit only* — no multi-network overlay, no Connector model. HelixVPN is self-hostable and adds the 1→N overlay.

4.2 Tailscale / Headscale — the coordination model

- **What it is:** a WireGuard mesh with a coordination server that pushes a desired-state "network map" (`MapResponse`) to each node, ACL-compiled topology, subnet routers, and (via Headscale) partial self-hosting [00 §6].
- **What HelixVPN borrows:** the **push-don't-poll coordination model** (P4) — the `WatchNetworkMap` server-stream is explicitly Tailscale-`MapResponse`-style [00 §8 P4], [SPEC §7.2]; the ACL → per-peer `AllowedIPs` compilation; and the **4via6** scheme for mapping overlapping advertised IPv4 LANs into a per-tenant ULA IPv6 space (D4) [00 §11 D4].
- **What HelixVPN diverges on:** Tailscale is a mesh (peers connect to peers); HelixVPN's MVP is a reverse-tunnel hub-and-spoke (edges dial the Gateway), with direct P2P + DERP-style relay deferred to Phase 2 (X4) [00 §9]. Tailscale lacks a full Mullvad-grade obfuscation stack.

4.3 WireGuard-based incumbents (NetBird, Netmaker, Firezone, wg-easy)

- **What they are:** self-hostable WireGuard control planes / mesh managers. The overview names **NetBird** as the closest OSS shape (WireGuard + management/signal server, self-hosted) [00 §6].

- **What HelixVPN borrows:** the architectural validation that a WG-crypto-core + a separate control/signal server is the right split — HelixVPN's P2 (WG is the crypto core, transports are pluggable beneath it) is the same instinct, made strict [00 §8 P2].
- **What HelixVPN diverges on:** these incumbents generally lack Mullvad-grade obfuscation (MASQUE/QUIC, DAITA), and none ship a first-party 8-platform app suite including Aurora + HarmonyOS from a shared codebase. HelixVPN treats obfuscation as a first-class pluggable layer, not an add-on.

> **UNVERIFIED** — the specific capability gaps attributed to NetBird/ Netmaker/ Firezone/wg-easy (e.g. "lacks MASQUE obfuscation", "no Aurora/HarmonyOS app") are competitive-product facts at a point in time. Re-verify in the §11.4.99 pass before external use (R1).

4.4 Cloudflare Tunnel + WARP — connector-dials-out + MASQUE

- **What it is:** an outbound-only connector (`cloudflared`) that dials the Cloudflare edge, plus WARP, a consumer client using MASQUE [00 §6].
- **What HelixVPN borrows:** the **connector-dials-out** model (the founding constraint, P3) and **MASQUE as a transport** [00 §6] (`[04_ARCH §0/§3.3]`).
- **What HelixVPN diverges on:** Cloudflare's coordination + edge are a proprietary global SaaS; HelixVPN is self-hostable, with the edge as a Rust component you run. HelixVPN's obfuscation is Mullvad-parity, broader than WARP.

4.5 Twingate / Zscaler — the ZTNA policy model

- **What they are:** enterprise Zero-Trust Network Access — per-user/per-app policy, no inbound exposure, identity-driven [00 §6].
- **What HelixVPN borrows:** the **ZTNA default-deny policy model** — need-to-know peer distribution, per-user ACL → reachability [00 §6], (S1/S3 in the security overview).
- **What HelixVPN diverges on:** these are SaaS, not self-hostable, and operate higher in the stack; HelixVPN is an L3 IP overlay with L4 port policy (it is explicitly NOT an L7 inspection appliance, §6) [00 §2.2].

4.6 Positioning matrix (from the overview)

This is the [00 §6] matrix, reproduced as the canonical positioning table. The non-HelixVPN cells carry the §4 `UNVERIFIED` blanket caveat.

System	Self-host	Obfuscation	Multi-network overlay	1st-party cross-platform apps	What HelixVPN borrows
Mullvad	☓ (service)	✅ best-in-class	☓ (privacy exit only)	✅	the obfuscation + privacy bar
Tailscale	partial (Headscale)	☓	✅ (mesh + subnet routers)	✅	the coordination / network-map model
NetBird	✅	limited	✅	✅	closest OSS shape (WG + mgmt/signal)
Cloudflare Tunnel + WARP	☓	MASQUE	✅	✅	connector-dials-out + MASQUE
Twingate / Zscaler	☓	n/a	✅ ZTNA	✅	the ZTNA policy model
HelixVPN	✅	✅ full Mullvad-parity stack	✅	✅ on 8 platforms, one codebase	the self-hosted *union* of all of the above

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5. Differentiators

> **Reconciled (§11.4.35, 2026-06-26):** the overview names **four** differentiators *to lead with externally* [00 §6] (`[04_ARCH §1.3]`); this section **expands them to seven** (D-1…D-7) for the internal positioning map. The "four" and the "seven" are not in conflict — four are the headline messaging set, seven are the full enumeration here.

The differentiators the overview tells us to lead with [00 §6] (`[04_ARCH §1.3]`) — **four to lead with externally, expanded to seven here** — each tied to the architecture that delivers it and the spec doc that owns it. None is aspirational; each maps to a concrete component.

D-1 — Self-hosted AND Mullvad-grade obfuscation (incl. MASQUE/QUIC)

The combination is the headline. Mullvad has the obfuscation but is a service; the self-hostable WG incumbents lack the obfuscation. HelixVPN ships both: a pluggable `Transport` layer beneath WireGuard (plain UDP, LWO, MASQUE/QUIC, Shadowsocks, UoT) selectable via an auto-escalation ladder [00 §9 F1–F8], [SPEC §5.2]. Owned by `../01-data-plane.md` (D1).

D-2 — Multi-network overlay: 1 user → N policy-scoped networks

The capability no incumbent self-hostable product combines with obfuscation: multiple Connectors each advertise their CIDRs, the Gateway compiles them into one *policed* overlay, and a single user reaches the granted subset [00 §4.2, §9 X1]. This implies three first-class problems the spec solves — overlapping RFC1918 ranges (D4 → 4via6), per-user ACL routing (default-deny → verdict maps), and split horizon (microsegmentation by default) [00 §4.2]. Owned by `../v02-data-plane/routing-and-addressing.md`

- `../v03-control-plane/svc-policy.md`.

D-3 — Outbound-only network onboarding (no inbound port-forward, ever)

The founding constraint, made a principle (P3): Connectors and Clients always dial the Gateway; no private network ever needs an inbound hole [00 §8 P3, §9 X2]. This removes the single largest attack surface (a public inbound port) and is what makes "expose my LAN without touching my router" literally true. Owned by the Connector spec + the security overview.

D-4 — One shared Rust + Flutter codebase across 8 platforms

The same Rust `helix-core` wraps obfuscation on the client and *unwraps* it on the edge (P5 — one implementation, three consumers); the same Flutter `helix-ui` builds all three apps via a flavor entrypoint [00 §8 P5, §5.2], [SPEC §3]. The platform reach — iOS, Android, **Aurora OS**, **HarmonyOS NEXT**, Windows, Linux, macOS, Web — is a differentiator no incumbent matches, with Aurora + HarmonyOS being unique [00 §9 X5]. Owned by `../03-client-core-and-ui.md` and Volume 4.

D-5 — Event-driven, sub-second control plane

State changes propagate as events; agents reconcile to a pushed desired-state with a **p99 < 1 s** convergence SLO, including policy edits and device revocation [00 §8 P4], [SPEC §8.1]. Polling/cron-restart loops cannot meet this. Owned by `../v03-control-plane/svc-coordinator.md`.

D-6 — No-logging-as-code (verifiable, not promised)

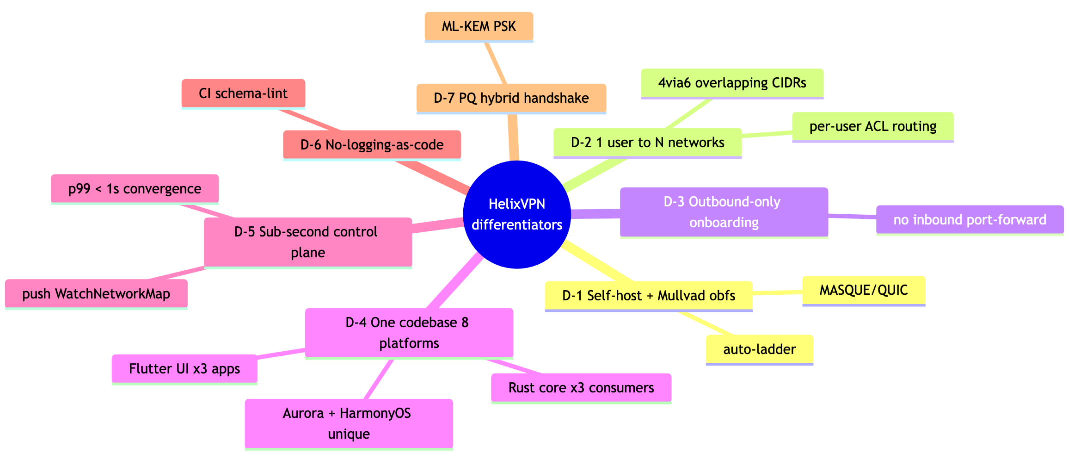
> **Reconciled (§11.4.35, 2026-06-26) — parity, not a Mullvad differentiator.** Mullvad already runs a strong no-logging stance, so "no-logging" is a **parity row**, not an edge over Mullvad. HelixVPN's genuine edge is that no-logging is **verifiable by construction** — you **self-host** the box and a CI schema-lint fails the build if a durable connection/traffic/packet table appears — rather than a trust-the-operator promise.

Privacy is a build property: no durable traffic table, CI-enforced (P7) [00 §8 P7]. The mechanical teeth are a CI schema-lint that fails the build on a forbidden durable table [00 §8 illustrative guard]. Owned by `../v05-security/no-logging-as-code.md`.

D-7 — Post-quantum-ready, hybrid-never-PQ-only

> **Reconciled (§11.4.35, 2026-06-26) — parity, not a Mullvad differentiator.** Mullvad ships post-quantum too, so a PQ handshake is a **parity row**, not an edge over Mullvad. HelixVPN's genuine edge is that the same hybrid PQ handshake runs on a stack **you self-host and can audit** — the differentiation is self-host + verifiability, not the PQ feature itself.

A PQ KEM (ML-KEM / FIPS-203) derives a PSK mixed into the classical WG handshake; an attacker must break both, and disabling PQ leaves classical WG secure [00 §9 F16], (S10). Owned by `../v05-security/post-quantum.md`.



6. Explicit non-goals

The negative space is load-bearing — it keeps every downstream document from drifting into an adjacent product. These are from [00 §2.2] and [SPEC §2], stated as binding product non-goals (not "later phases").

Non-goal	Why excluded	Where enforced
NOT a forked/re-rolled WireGuard crypto	the crypto core is the audited WG Noise IK construction; HelixVPN changes only how encrypted datagrams *look on the wire*	`../01-data-plane.md`; CI lint forbids touching WG crypto primitives [00 §2.2]
NOT a separate "QUIC protocol" alongside WG	Mullvad's QUIC mode *is* WG-over-MASQUE/HTTP-3, not a different protocol	D1; transport spec `01` (`[research-masque]`)
NOT a full system VPN in a browser	browsers cannot open a TUN device; Web = Console (management) + an *optional* in-page WASM MASQUE proxy that proxies the **browser's own** traffic only	`../v04-client/web-console.md`; §10 scope [00 §2.2]
NOT a logging / lawful-intercept / DLP appliance	no-logging is an architectural build property, not a toggle; *control* actions are audited, *traffic* never is	data-model spec `02`; CI schema-lint [00 §2.2]
NOT a generic SDN / service-mesh / L7 API gateway	HelixVPN operates at L3 (IP overlay) + L4 port policy; it is not Envoy/Istio and does not terminate application TLS for inspection	policy spec; §2.1 scope [00 §2.2]
NOT a consumer "free VPN" ad-funded service	the default deployment is self-hosted; a managed offering is an optional later SKU, not the reason to exist	§7 [00 §2.2]
NOT dependent on inbound port-forwarding on any joined network	the reverse-tunnel insight is foundational (P3)	Principle 3; Connector spec [00 §2.2]

6.1 Two honesty markers (hard physical limits, stated to users)

Per §11.4.6, two product claims have *hard physical limits* and MUST be stated to users without hand-waving [00 §2.2] (`[04_ARCH §5.7]`):

1. **"Fully responsive web app" ≠ system VPN.** The Web build is the Console plus a browser-scoped proxy; it cannot tunnel the whole OS (no TUN in a browser).
2. **iOS Network Extension ~15 MB working-set ceiling.** A measured historical constraint that shapes the core-language decision (D2) and is verified on-device in Phase-0 gate G3, never assumed [00 §2.2], [SPEC §8.0 G3].

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7. Positioning statement & messaging spine

7.1 The positioning statement (for-…-who-…-is-…-unlike)

> **For** self-hosters, home-lab operators, and small MSPs **who** need both Mullvad-grade privacy and Tailscale-grade multi-network coordination without a SaaS dependency, **HelixVPN is** a self-hostable overlay network with a privacy-VPN front end that gives one user policy-scoped access to many private networks — without any inbound port-forward — on 8 platforms from one codebase. **Unlike** Mullvad (a service, privacy-exit only), Tailscale (no full obfuscation), or the self-hostable WG incumbents (no Mullvad-grade obfuscation), HelixVPN is the self-hosted *union* of all of them. [00 §6], [SPEC §2]

7.2 Per-audience messaging

Audience	Lead message	Proof point
Privacy consumer	"Mullvad's privacy stack, on a box *you* own — verifiable no-logs, not promised no-logs"	D-1, D-6
Home-lab operator	"Reach your lab from anywhere without opening a single router port"	D-2, D-3
Small MSP / business admin	"One overlay, N client networks, per-user default-deny policy, control-action audit"	D-2, D-5
Platform-diverse org	"One app suite on iOS, Android, Aurora, HarmonyOS, Windows, Linux, macOS, Web"	D-4

> **UNVERIFIED** — buyer/audience segmentation and the relative appeal of each message are go-to-market hypotheses, not architectural facts. They are recorded as the messaging spine, not as requirements.

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8. Strategic risks to the positioning

Honest disclosure per §11.4.6/§11.4.118 — the positioning rests on assumptions that can fail; each is a tracked risk, not a silent confidence.

Risk	Description	Mitigation / where tracked
R-V1	The iOS NE ~15 MB memory ceiling could make the shared-Rust-core client strategy infeasible on iOS, undermining D-4 (one codebase)	Phase-0 gate **G3** is make-or-break and decides D2; failure re-opens the client strategy [SPEC §8.0]
R-V2	MASQUE may not survive real DPI at acceptable throughput, undermining D-1	Phase-0 gate **G2** (MASQUE through a DPI block ⩾50% of plain); failure re-opens D1 [00 §11 D1]
R-V3	Aurora + HarmonyOS native tunnel-shim work is the "biggest platform risk" — the unique D-4 reach could slip to Phase 3 or fail	scoped to Phase 3 with on-device exit gates [00 §10.4], [SPEC §8.3]
R-V4	The "no self-hostable product ships 1→N + obfuscation today" claim (D-2) may be eroded by a competitor	UNVERIFIED-tagged; re-verified each §11.4.99 pass (R1)
R-V5	Licensing (D8) unresolved could fork the community vs the managed-SKU business	OWNER-GATED decision before public release; `LICENSE.PENDING` until then [00 §7.2]

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9. Cross-document contracts this document fixes

This document does not invent product scope — it *positions* the scope fixed by the overview and spine. The contracts it relies on and forwards:

1. **The four-lineage union pitch** (§1) → inherited from [00 §1], [SPEC §1]; every differentiator (§5) traces to a borrowed lineage.

2. **The dual consumer+business model** (§3) → the same Gateway/core serves privacy-exit and overlay-access; consumed by `personas-and-roles.md` (which enumerates the resulting personas) and `functional-requirements.md` (which numbers the resulting requirements).
3. **The seven non-goals** (§6) → binding boundaries from [00 §2.2]; every FR in `functional-requirements.md` MUST stay inside them.
4. **Differentiators map to owning specs** (§5) → each D-n names the Volume 2–5 doc that delivers it; the requirements traceability matrix (`../v00-meta/requirements-traceability.md`, planned) closes the loop.
5. **Open decisions stay open** (D1/D2/D4/D8 referenced) → never silently resolved here; they live in [00 §11] / [SPEC §9].

Sources verified

- `00-product-scope-and-principles.md` §1 (product definition), §2 (is/is-not + honesty markers), §6 (prior art + differentiators), §7 (hosting/licensing), §8 (principles), §9 (parity matrix), §10 (phase scope), §11 (decisions D1–D7) — primary spine for this document.
- `SPECIFICATION.md` §1 (executive summary), §2 (product vision), §3 (roles/apps), §8 (roadmap/gates), §9 (decision register incl. D8).
- `04-security-privacy-pki.md` §0.1 (S6/S7 no-logging + control-only audit) — the privacy/no-logging product promise.
- `MASTER_INDEX.md` Volume 1 row (this document's declared scope: "Vision, market, prior-art deep-dive, differentiators, non-goals").
- Primary-source chain preserved from the overview: `[04_ARCH §0/§1/§2/§3.3/§5.7]`, `[05_YBO]`, `[research-masque]` (RFC 9298/9297/9221/9484).

Honesty note (§11.4.6): every competitive-market claim (prior-art capability gaps, "no product ships X today", category-convergence trend, audience segmentation) is marked `UNVERIFIED` and deferred to the §11.4.99 latest-source verification pass (REFINEMENT_NOTES R1). Architectural claims about HelixVPN are cited to the overview/spine. No competitor capability was asserted from training memory.

Constitution bindings: §11.4.44 (revision header), §11.4.6 (no-guessing — UNVERIFIED markers on all market claims), §11.4.66 (open decisions carry options + recommendation), §11.4.99 (latest-source re-verification deferred + tracked), §11.4.35 (inherits, never weakens, the overview's invariants), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).